Laws of Indices

Exam code: 9MA0

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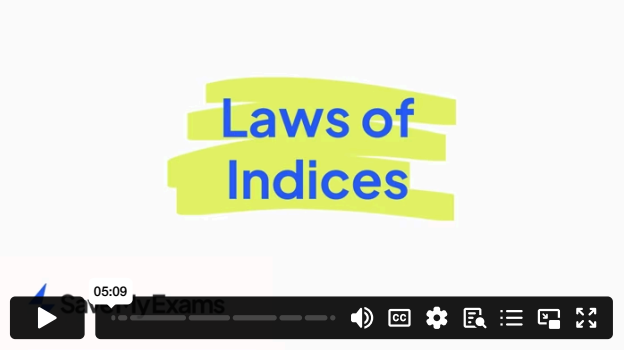

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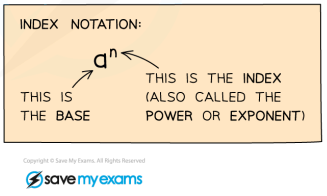
Help me understand

Laws of indices

What do I need to know about laws of indices?

- Laws of indices (or index laws – 'index' is the singular, 'indices' is the plural) allow you to simplify and manipulate expressions involving powers
- The index laws you need to know and use are summarised here:

- $a^m \times a^n = a^{m+n}$
- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $(ab)^n = a^n b^n$
- $a^1 = a$
- $a^0 = 1$
- $a^{\frac{1}{m}} = \sqrt[m]{a}$
- $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
- $a^{-m} = \frac{1}{a^m}$



How do I change the base of a term?

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- For example $9^4 = (3^2)^4 = 3^{2\times 4} = 3^8$
- Using the above can then help with problems like $9^4 \div 3^7 = 3^8 \div 3^7 = 3^{8-7} = 3^1 = 3$

Examiner Tips and Tricks

- Index laws only work with terms that have the same base, so something like $2^3 \times 5^2$ cannot be simplified using index laws

Worked Example

Given that $y = \frac{1}{27}x^3$ express each of the following in the form kx^n, where k and n are constants.

a) $y^{\frac{1}{3}}$

b) $\frac{1}{9}y^{-2}$

a) $y^{\frac{1}{3}} = \left(\frac{1}{27}x^3\right)^{\frac{1}{3}}$

$= \left(\frac{1}{27}\right)^{\frac{1}{3}} (x^3)^{\frac{1}{3}}$

$= \sqrt[3]{\frac{1}{27}}\, x^{3\times\frac{1}{3}}$

$= \frac{1}{3}x^1 = \frac{1}{3}x$

b) $\frac{1}{9}y^{-2} = \frac{1}{9}\left(\frac{1}{27}x^3\right)^{-2}$

$= \frac{1}{9}\left(\frac{1}{27}\right)^{-2}(x^3)^{-2}$

$= \frac{1}{9}\left(\frac{1}{\frac{1}{27}}\right)^2 x^{-6}$

$= \frac{1}{9}(27)^2 x^{-6}$

$= 81x^{-6}$

$= \frac{81}{x^6}$

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